Ultra-sparse MoE

Hundreds of narrow routed experts plus one shared expert

DeepSeek-AI · DeepSeek-V4 (also Kimi K2, GLM-5)

Explainer deck · Industrial Pretraining Architecture & Infra Tech Landscape (2026)

MoE & Routing · arxiv.org/abs/2606.19348

Frontier open MoEs share one FFN shape

Model	Total / active	Routed experts	Routed / token	Shared
DeepSeek-V3 (2024)	671B / 37B	256	8	1
GLM-5	744B / 40B	**256**	**8**	**1**
Kimi K2	1.04T / 32.6B	**384**	**8**	**1**
DeepSeek-V4-Flash	284B / 13B	**256**	**6**	**1**
DeepSeek-V4-Pro	1.6T / 49B	**384**	**6**	**1**

Hundreds of routed experts, 6–8 per token, plus one always-on shared expert

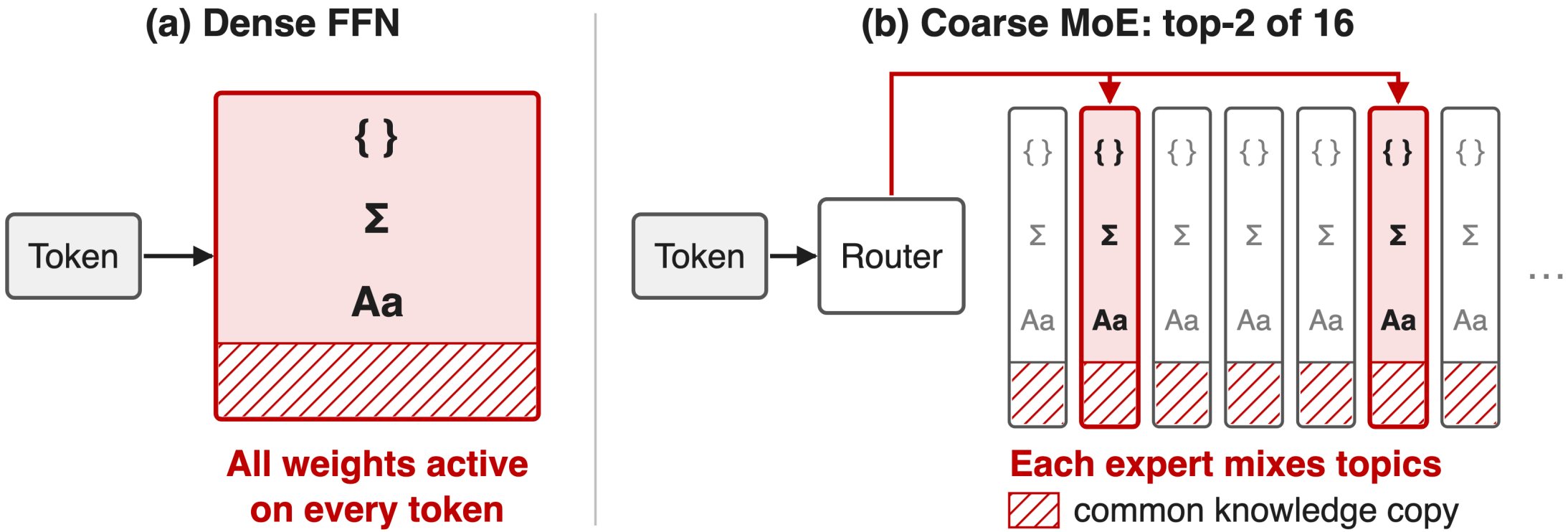
Coarse experts cost 1.5× the compute

1.5×

the expert parameters and FLOPs a classic top-2 MoE (GShard 2.9B) spends to match a 2B fine-grained MoE with a shared expert

DeepSeekMoE, arXiv 2401.06066, Abstract and §1

Dense FFNs and coarse MoEs both waste FLOPs



Few big experts: each mixes topics, and all relearn common knowledge

Insight: at equal FLOPs, more experts win

- Kimi K2 sweep: **8** active + **1** shared expert, vary the total
- Train and val. loss fall consistently, sparsity **8** → **64**

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Total parameters buy quality; activated parameters set the cost

A V4-Pro token uses 7 of 385 experts



Per layer, a token touches 1.8% of the expert weights (derived)

❶ Narrow experts multiply the combinations

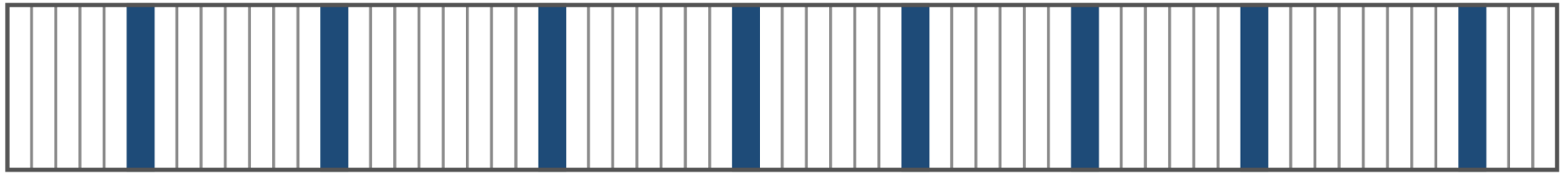
16 experts
width f · top-2



120 possible expert sets

split ×4

64 experts
width f/4 · top-8



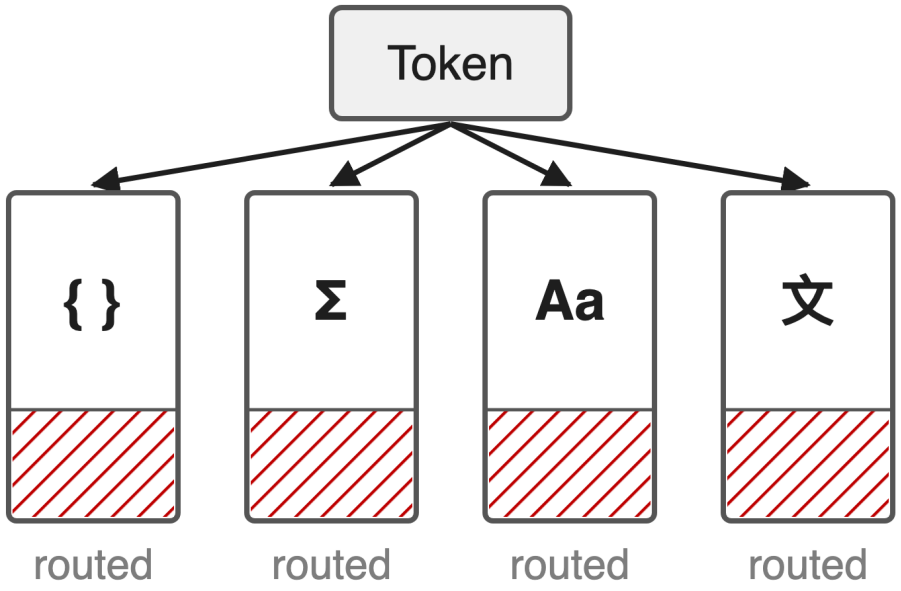
4,426,165,368 possible expert sets

same active width: 2f = 8 · f/4

Narrow experts let the router compose knowledge per token

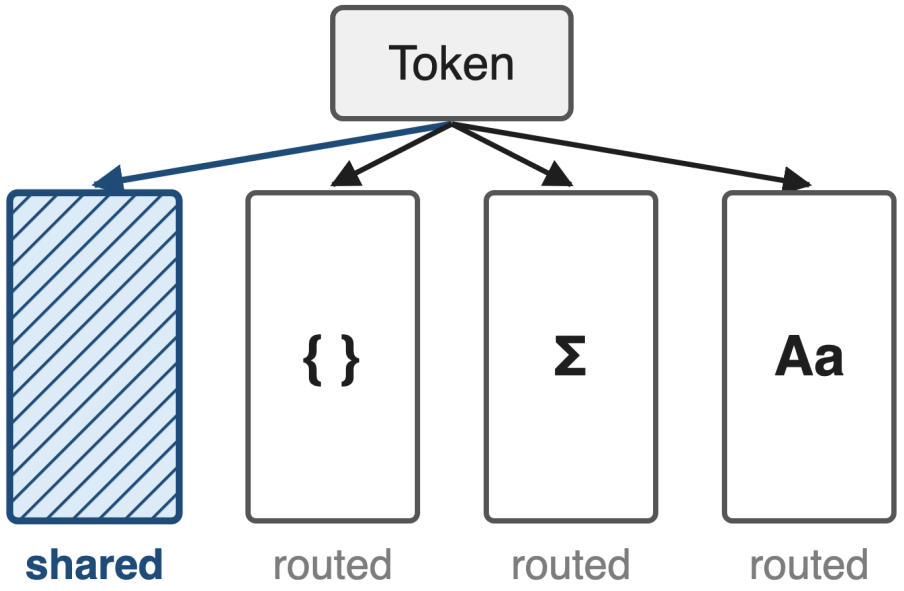
❷ One shared expert holds common knowledge

(a) Routed only: top-4



common knowledge × 4

(b) 1 shared + top-3 routed

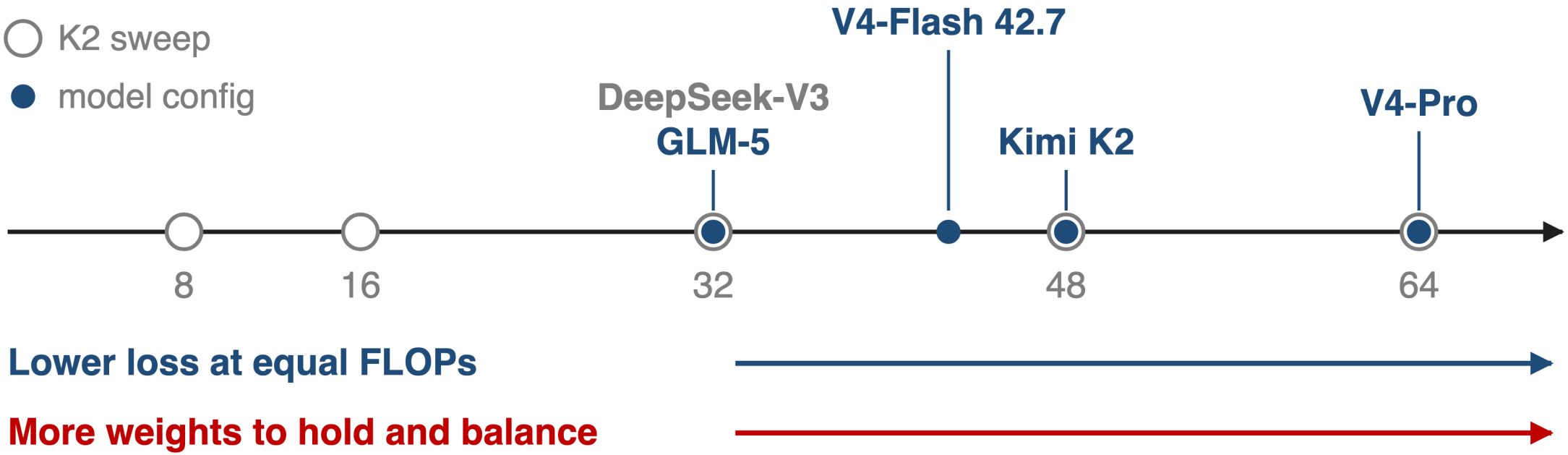


common knowledge × 1

Routed top-k drops by one, so per-token FLOPs stay fixed

❸ Sparsity is a dial capped by infrastructure

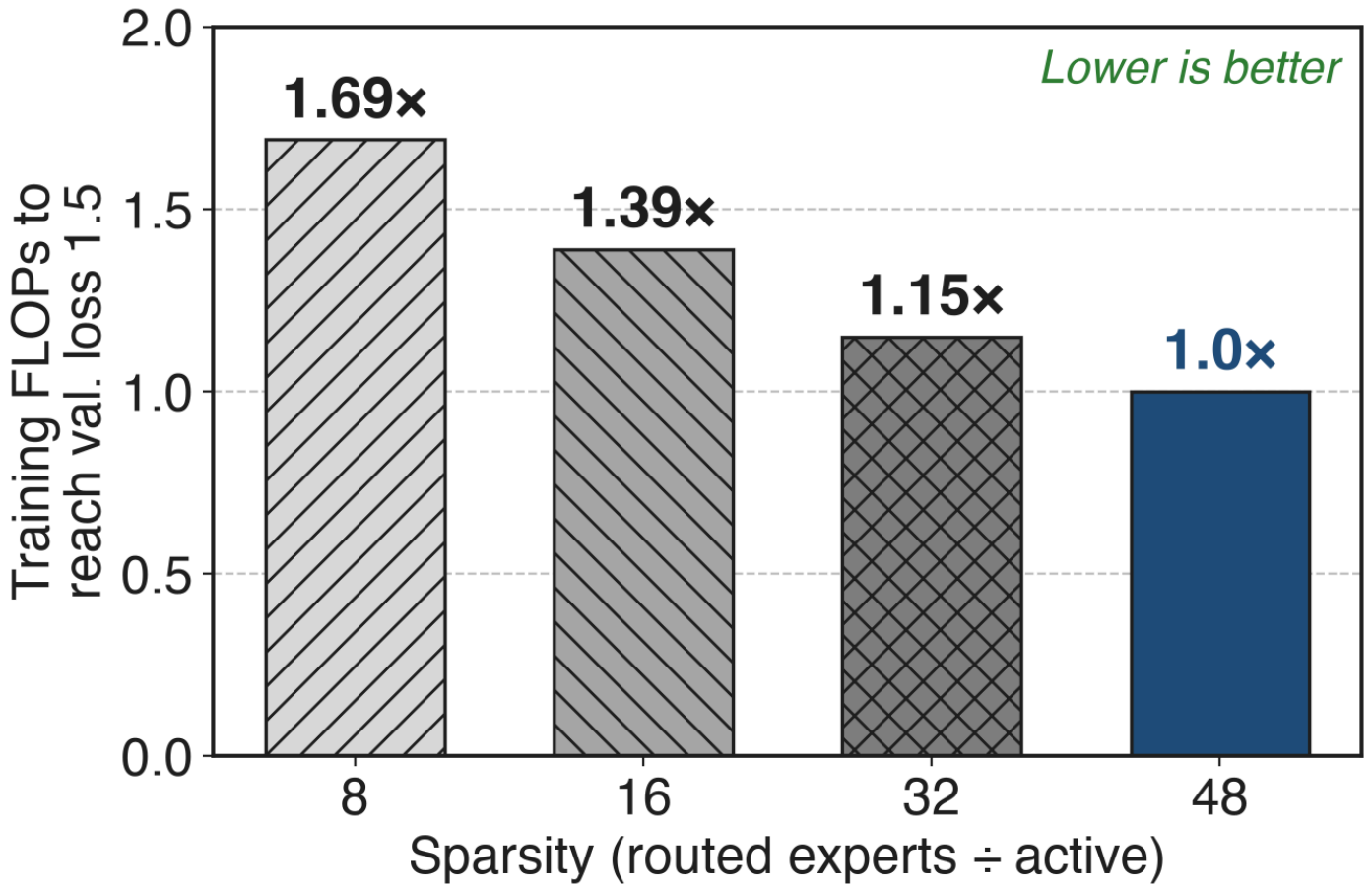
sparsity = routed experts ÷ routed experts per token



K2 stops at 48 for infra cost; V4-Pro runs at 64 (derived)

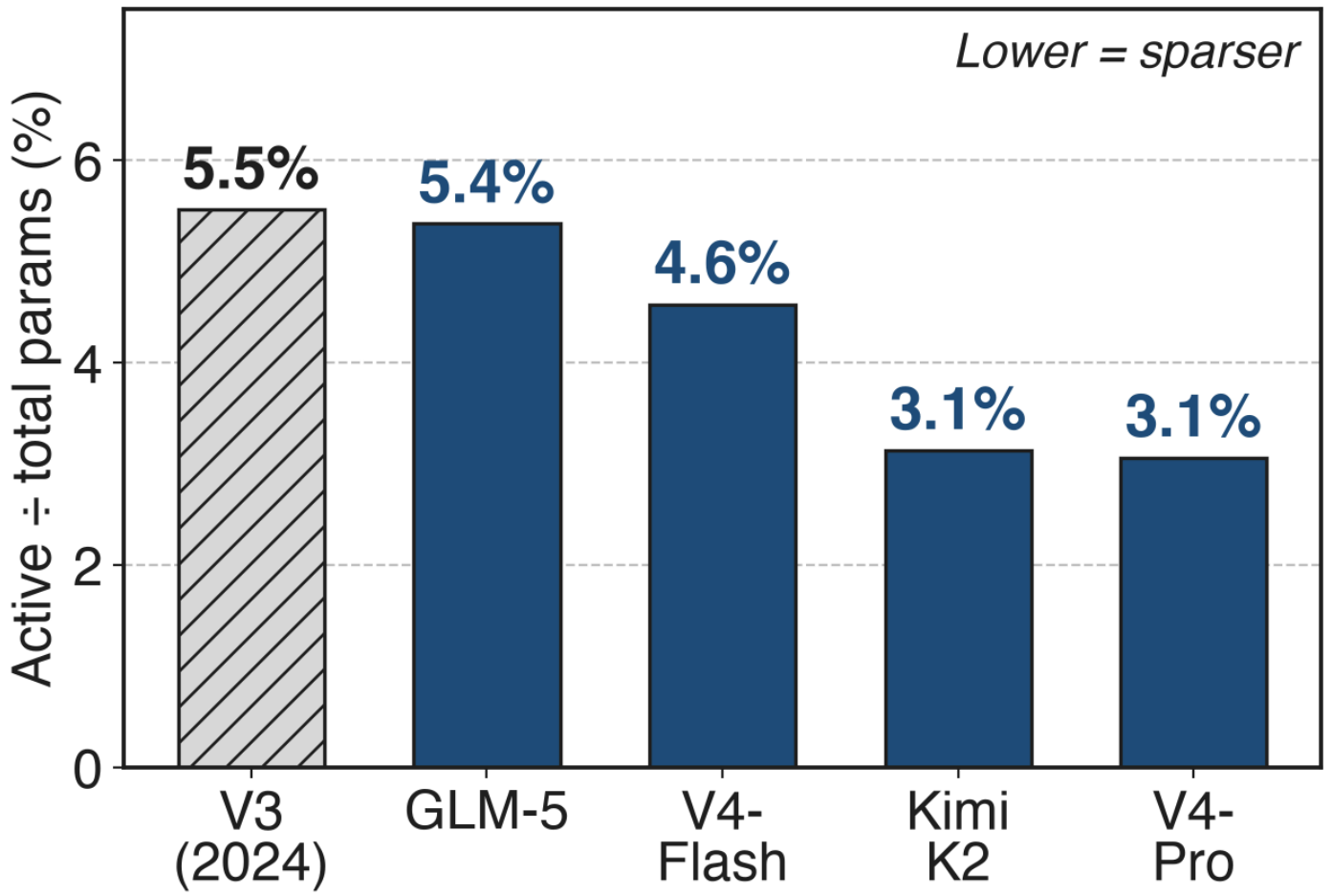
Sparsity 48 needs 1.69× fewer FLOPs than 8

- Kimi K2 sweep, same val. loss **1.5**
- **8** active + **1** shared expert in every run
- Gain shrinks: **1.15×** over sparsity 32



Frontier MoEs activate only 3–5% of weights

- Experts hold **~97%** of V4-Pro's **1.6T** weights
- Each token uses **~28B** of them (derived)



The cost: memory, all-to-all, balance

- **All experts resident**: K2 needs **~6 TB** for weights + grads
- **All-to-all every layer**: GLM-5 trims depth; K2 keeps EP **16**
- **Balance and spikes**: loss-free bias; MoE outliers spike the loss
- K2 stops at sparsity **48**

Sparsity moves cost from FLOPs to memory and network

Summary

- Coarse MoE spends **1.5×** compute for equal quality
- **Insight**: at equal FLOPs, more and smaller experts lower loss
- Narrow routed experts + **1** shared expert; sparsity **32–64**
- **1.69×** fewer FLOPs (sparsity 48 vs 8); **3–5%** of weights active

https://arxiv.org/abs/2606.19348